

Task 1 (22 Points)

- (3 P) * a) New slack variable $z \geq 0 \Rightarrow \bar{\mathbf{x}} = [\mathbf{x}^T, z]^T$
leads to the standard form

$$\begin{aligned} \min_{\bar{\mathbf{x}} \in \mathbb{R}^5} \quad & [15 \ 8 \ 6 \ -1 \ 0] \bar{\mathbf{x}} \\ \text{s.t.} \quad & \begin{bmatrix} 3 & 2 & 0 & -1 & 0 \\ 3 & 1 & 2 & -1 & -1 \end{bmatrix} \bar{\mathbf{x}} = \begin{bmatrix} 1.5 \\ 1 \end{bmatrix}, \quad \bar{\mathbf{x}} \geq \mathbf{0} \end{aligned}$$

- (7 P) b) Lagrangian function

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = (\mathbf{A}^T \boldsymbol{\lambda} - \boldsymbol{\mu} + \mathbf{c})^T \mathbf{x} - \boldsymbol{\lambda}^T \mathbf{b}$$

The dual problem is

$$\begin{aligned} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} \quad & \theta(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \begin{cases} -\boldsymbol{\lambda}^T \mathbf{b} & \text{if } \mathbf{A}^T \boldsymbol{\lambda} - \boldsymbol{\mu} + \mathbf{c} = \mathbf{0} \\ -\infty & \text{else,} \end{cases} \\ \text{s.t.} \quad & \boldsymbol{\mu} \geq \mathbf{0}. \end{aligned}$$

or

$$\begin{aligned} \max_{\boldsymbol{\lambda} \in \mathbb{R}^2} \quad & -\boldsymbol{\lambda}^T \mathbf{b} \\ \text{s.t.} \quad & \mathbf{A}^T \boldsymbol{\lambda} + \mathbf{c} \geq \mathbf{0} \end{aligned}$$

which leads to

$$\begin{aligned} \max_{\boldsymbol{\lambda} \in \mathbb{R}^2} \quad & -1.5\lambda_1 - \lambda_2 \\ \text{s.t.} \quad & 3\lambda_1 + 3\lambda_2 + 15 \geq 0 \\ & 2\lambda_1 + \lambda_2 + 8 \geq 0 \\ & 2\lambda_2 + 6 \geq 0 \\ & -\lambda_1 - \lambda_2 - 1 \geq 0 \\ & -\lambda_2 \geq 0 \end{aligned}$$

- (7 P) c) Graphical solution with

- Contour lines: $\lambda_2 = -1.5\lambda_1 - c$
- Constraints:

- 1) $\lambda_2 \geq -\lambda_1 - 5$
- 2) $\lambda_2 \geq -2\lambda_1 - 8$
- 3) $\lambda_2 \geq -3$
- 4) $\lambda_2 \leq -\lambda_1 - 1$
- 5) $\lambda_2 \leq 0$

$$\Rightarrow \lambda^* = [-3, -2]^T$$

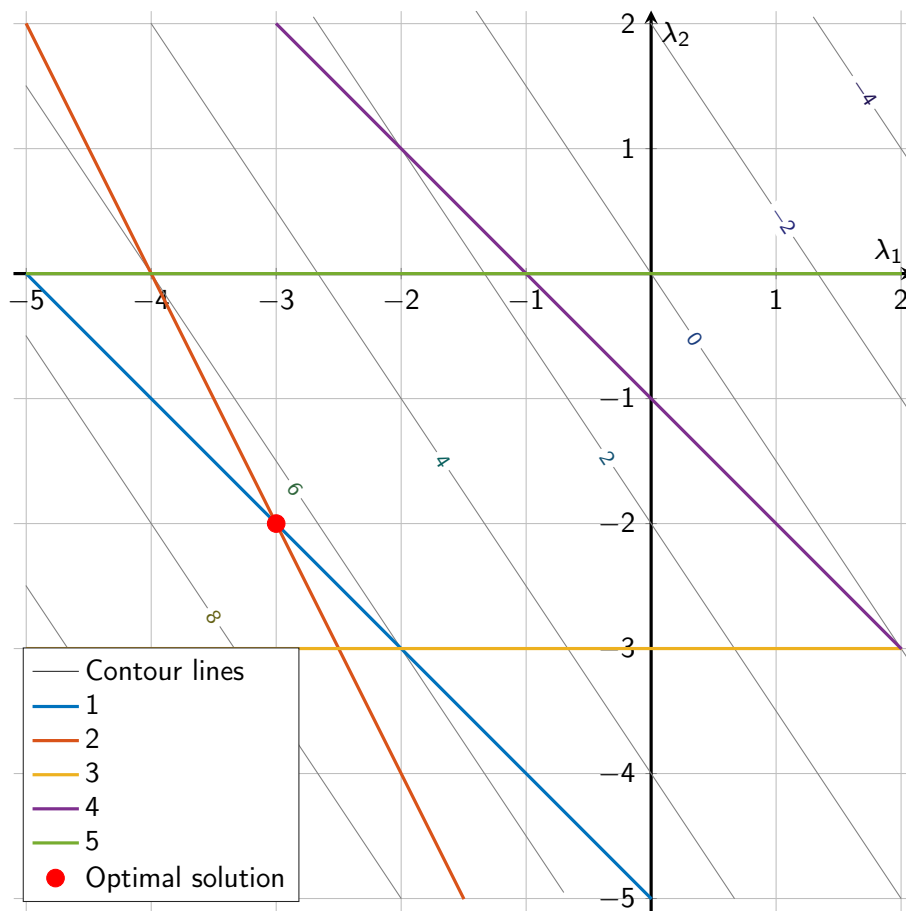


Figure 1: Contourlines and admissible set for Task 1.

(5 P) d) Calculation of the primal solution using the dual solution:

- Calculation of μ^* : $A^T \lambda^* - \mu^* + c = 0 \Rightarrow \mu^* = A^T \lambda^* + c$

$$\mu^* = \begin{bmatrix} -9 - 6 \\ -6 - 2 \\ 0 - 4 \\ 3 + 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 15 \\ 8 \\ 6 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \\ 4 \\ 2 \end{bmatrix}$$

- Since $\mu_{3,4,5}^* \neq 0$ for the optimal primal solution $x_{3,4}^* = 0$ and $z^* = 0$ must hold. The remaining solution can be determined via the two equality constraints $Ax^* = b$ of the primal problem.

$$3x_1^* + 2x_2^* = 1.5, \quad 3x_1^* + x_2^* = 1$$

$$\Rightarrow x^* = \left[\frac{1}{6}, \frac{1}{2}, 0, 0, 0 \right]^T$$

- Duality gap: $c^T x^* = \frac{15}{6} + 4 = \frac{13}{2}$ and $-\lambda^* b = \frac{9}{2} + 2 = \frac{13}{2} \Rightarrow$ No duality gap.

Task 2 (25 Points)

(4P) *a) Lagrangian function

$$L(\mathbf{x}, \boldsymbol{\mu}) = 2(x_1 - t)^2 + x_2^2 + \mu_1(2x_1 - x_2 - 4) + \mu_2(-x_1 - x_2 + 3) + \mu_3(x_2 - 4) + \mu_4(-x_1)$$

Optimality conditions

$$\begin{aligned}\nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\mu}) &= \begin{bmatrix} 4(x_1 - t) + 2\mu_1 - \mu_2 - \mu_4 \\ 2x_2 - \mu_1 - \mu_2 + \mu_3 \end{bmatrix} = \mathbf{0} \\ h_i(\mathbf{x}) &\leq 0, \quad i = 1, \dots, 4 \\ \mu_i &\geq 0, \quad i = 1, \dots, 4 \\ h_i(\mathbf{x})\mu_i &= 0, \quad i = 1, \dots, 4\end{aligned}$$

(6P) b) Plug in $\mathbf{x}^* = [\frac{7}{3}, \frac{2}{3}]^T$ in inequality constraints:

$$\begin{aligned}h_1(\mathbf{x}) &= \frac{14}{3} - \frac{2}{3} - 4 = 0 \Rightarrow \text{active: } \mu_1 \geq 0 \\ h_2(\mathbf{x}) &= -\frac{7}{3} - \frac{2}{3} + 3 = 0 \Rightarrow \text{active: } \mu_2 \geq 0 \\ h_3(\mathbf{x}) &= \frac{7}{3} - 4 \leq 0 \Rightarrow \text{inactive: } \mu_3 = 0 \\ h_4(\mathbf{x}) &= -\frac{7}{3} \leq 0 \Rightarrow \text{inactive: } \mu_4 = 0\end{aligned}$$

For $\nabla_{\mathbf{x}} L(\mathbf{x}^*, \boldsymbol{\mu})$ it follows that

$$4(\frac{7}{3} - t) + 2\mu_1 - \mu_2 = 0 \tag{8a}$$

$$\frac{4}{3} - \mu_1 - \mu_2 = 0 \Rightarrow \mu_1 = \frac{4}{3} - \mu_2 \tag{8b}$$

Inserting (8b) in (8a) gives

$$4\frac{7}{3} - 4t + 2\frac{4}{3} - 3\mu_2 = 0 \Rightarrow t = 3 - \frac{3}{4}\mu_2.$$

Since $\mu_2 \geq 0$ it follows that $t \leq 3$. Additionally, from (8b) and $\mu_1 \geq 0$ also follows

$$\mu_2 \leq \frac{4}{3}$$

which leads to $t \geq 2$. $\Rightarrow$ For $t \in [2, 3]$ the point $\mathbf{x}^*$ satisfies the optimality conditions.

(4 P) *c) Constraints:

- 1) $x_2 \geq 2x_1 - 4$
- 2) $x_2 \geq -x_1 + 3$
- 3) $x_2 \leq 4$
- 4) $x_1 \geq 0$

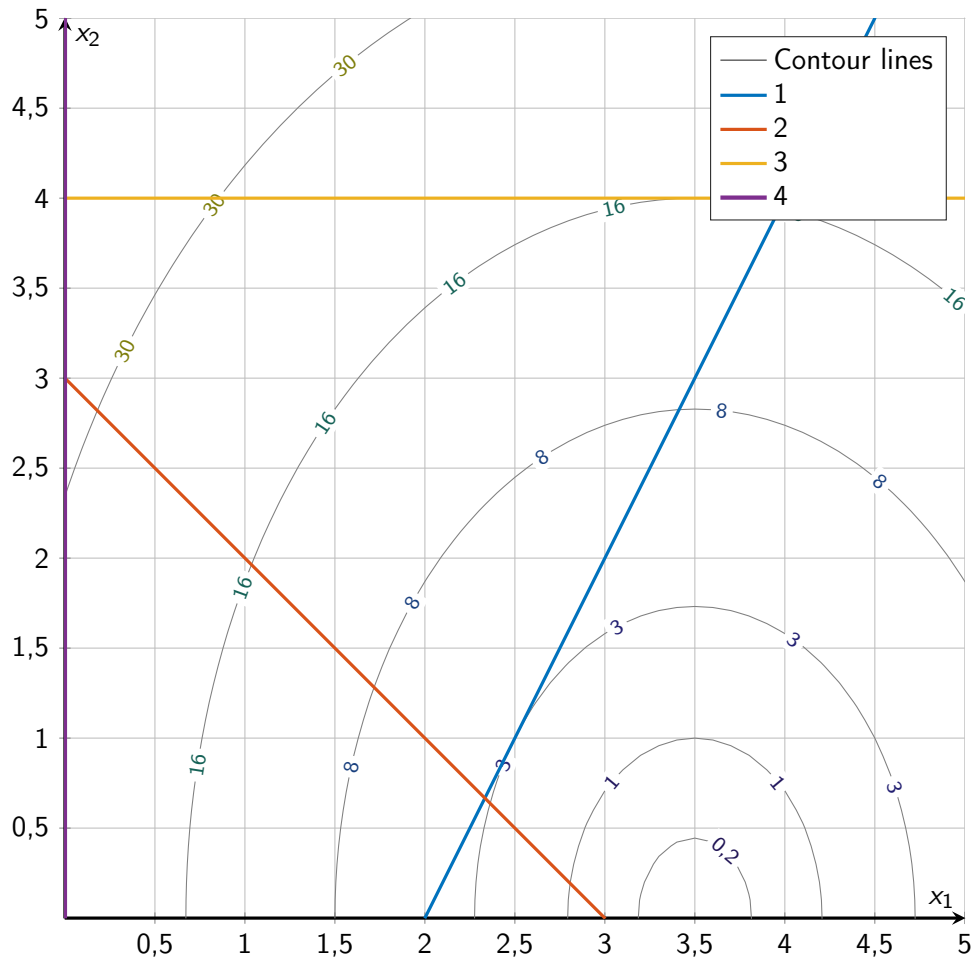


Figure 2: Admissible set for Task 2.

It is a convex problem, since the cost function is a quadratic positive-definite function (convex), and the admissible set is convex as well.

(11 P) d) Solution with active set method:

- $\mathbf{x}^0 = [0, 4]^T \Rightarrow \mathcal{A}^0 = \{3, 4\}$
Calculate multipliers from

$$\nabla_{\mathbf{x}} L(\mathbf{x}^0, \boldsymbol{\mu}) = \begin{bmatrix} -14 - \mu_4 \\ 8 + \mu_3 \end{bmatrix} = \mathbf{0} \Rightarrow \mu_4 = -14, \mu_3 = -8$$

thus deactivate 4

- $\mathcal{A}^1 = \{3\}$ and graphically minimizing problem leads to $\mathbf{x}^1 = [3.5, 4]^T$

Calculate multipliers

$$\nabla_x L(\mathbf{x}^1, \boldsymbol{\mu}) = \begin{bmatrix} 0 \\ 8 - \mu_3 \end{bmatrix} = \mathbf{0} \Rightarrow \mu_3 = -8$$

thus deactivate 3

- $\mathcal{A}^2 = \{\}$ and graphically minimizing problem leads to $\mathbf{x}^2 = [3.5, 3]^T \Rightarrow$ Constrain 1 gets active
- $\mathcal{A}^3 = \{1\}$ and graphically minimizing problem leads to $\mathbf{x}^3 = [2.5, 1]^T$
Berechnen der Multiplikatoren aus

$$\nabla_x L(\mathbf{x}^3, \boldsymbol{\mu}) = \begin{bmatrix} 4 + 2\mu_1 \\ 2 - \mu_1 \end{bmatrix} = \mathbf{0} \Rightarrow \mu_1 = 2$$

thus $\mathbf{x}^3 = \mathbf{x}^*$.

Task 3 (25 Points)

(3 P) *a) Hamiltonian function:

$$H(x, u, \lambda) = l(x, u) + \lambda f(x, u) = \frac{1}{a}(2-a)x^2 + \frac{1}{a}u^a + \lambda(u-x)$$

First-order optimality conditions:

$$\dot{x} = \frac{\partial H}{\partial \lambda} = u - x, \quad x(0) = 0$$

$$\dot{\lambda} = -\frac{\partial H}{\partial x} = -\frac{2}{a}(2-a)x + \lambda$$

$$0 = \frac{\partial H}{\partial u} = u^{a-1} + \lambda$$

$$\lambda(1) = \frac{\partial V}{\partial x}(x(1), 1) = x(1)$$

(4 P) b) Singular case, if H_u is independent of u . Thus the singular case occurs for $a = 1$.
Determine degree of singularity:

$$H_u = 1 + \lambda$$

$$\frac{\partial H_u}{\partial t} = \dot{\lambda} = -2x + \lambda$$

$$\frac{\partial^2 H_u}{\partial t^2} = -2\dot{x} + \dot{\lambda} = -2(u-x) + (-2x + \lambda) = -2u + \lambda$$

$\Rightarrow$ Degree of singularity $r = 2$

(3 P) *c) Hamiltonian function:

$$H(x, u, \lambda) = \frac{1}{2}u^2 + \lambda_1 x_2 + \lambda_2 u$$

First-order optimality conditions:

$$\dot{x}_1 = x_2, \quad x_1(0) = 1, \quad x_1(1) = 0$$

$$\dot{x}_2 = u, \quad x_2(0) = 0$$

$$\dot{\lambda}_1 = 0$$

$$\dot{\lambda}_2 = -\lambda_1, \quad \lambda_2(1) = x_2(1)$$

$$0 = u + \lambda_2$$

(12 P) d) Determination of the optimal trajectories from the optimality conditions

$$\begin{aligned}
 \dot{\lambda}_1 &= 0 & \Rightarrow & \lambda_1 = c_1 \\
 \dot{\lambda}_2 &= -\lambda_1 & \Rightarrow & \lambda_2 = -c_1 t + c_2 \\
 u &= -\lambda_2 = c_1 t - c_2 \\
 \dot{x}_2 &= u = c_1 t - c_2 & \Rightarrow & x_2 = \frac{1}{2} c_1 t^2 - c_2 t + c_3 \\
 \dot{x}_1 &= x_2 & \Rightarrow & x_1 = \frac{1}{6} c_1 t^3 - \frac{1}{2} c_2 t^2 + c_3 t + c_4
 \end{aligned}$$

Calculation of the constants c_i from the boundary constraints:

$$\begin{aligned}
 x_1(0) &= c_4 = 1 \\
 x_2(0) &= c_3 = 0 \\
 x_1(1) &= \frac{1}{6} c_1 - \frac{1}{2} c_2 + 1 \stackrel{!}{=} 0 \\
 \lambda_2(1) &= -c_1 + c_2 \stackrel{!}{=} x_2(1) = \frac{1}{2} c_1 - c_2 \Rightarrow c_2 = \frac{3}{4} c_1 \\
 \Rightarrow \frac{1}{6} c_1 - \frac{3}{8} c_1 + 1 &= -\frac{5}{24} c_1 + 1 \stackrel{!}{=} 0 \Rightarrow c_1 = \frac{24}{5}, c_2 = \frac{18}{5}
 \end{aligned}$$

Optimal solution:

$$\begin{aligned}
 x_1^* &= \frac{4}{5} t^3 - \frac{9}{5} t^2 + 1 \\
 x_2^* &= \frac{12}{5} t^2 - \frac{18}{5} t \\
 u^* &= \frac{24}{5} t - \frac{18}{5} \\
 \lambda_1^* &= \frac{24}{5} \\
 \lambda_2^* &= -\frac{24}{5} t + \frac{18}{5}
 \end{aligned}$$

(3 P) *e) Consideration of input constraints:

$$u^* = \arg \min_{u \in [u_{\min}, u_{\max}]} H = \arg \min_{u \in [u_{\min}, u_{\max}]} \frac{1}{2} u^2 + \lambda_2 u$$

Calculate minimum:

$$\frac{\partial H}{\partial u} = u + \lambda_2 \stackrel{!}{=} 0 \Rightarrow u^0 = -\lambda_2 \quad (\text{as solution from 3d})$$

$$\Rightarrow u^* = \begin{cases} u_{\min}, & \text{if } u^0 < u_{\min} \\ u^0, & \text{if } u^0 \in [u_{\min}, u_{\max}] \\ u_{\max}, & \text{if } u^0 > u_{\max} \end{cases}$$

Task 4 (18 Points)

(6 P) * a) Newton method:

- Quadratic function: $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{b}^T \mathbf{x} + \mathbf{c}$, $\mathbf{Q} = \text{pos. def.}$

- Gradient and Hessian $\nabla f(\mathbf{x}) = \mathbf{Q} \mathbf{x} + \mathbf{b}$, $\nabla^2 f(\mathbf{x}) = \mathbf{Q}$

$\Rightarrow$ unique global minimum at $\mathbf{x}^* = -\mathbf{Q}^{-1} \mathbf{b}$

- Start at $\mathbf{x}^0$ in direction $\mathbf{s}^0 = -[\nabla^2 f(\mathbf{x}^0)]^{-1} \nabla f(\mathbf{x}^0) = -\mathbf{Q}^{-1}(\mathbf{Q} \mathbf{x}^0 + \mathbf{b})$:

$$\begin{aligned} \mathbf{x}^1 &= \mathbf{x}^0 + \mathbf{s}^0 \quad (\text{full step size } \alpha^0 = 1) \\ &= \mathbf{x}^0 - \mathbf{Q}^{-1}(\mathbf{Q} \mathbf{x}^0 + \mathbf{b}) \\ &= -\mathbf{Q}^{-1} \mathbf{b} = \mathbf{x}^* \end{aligned}$$

$\Rightarrow$ Convergence after one step for all $\mathbf{x}^0$

Conjugate gradient method with exact line search needs max. n iterations.

(8 P) * b) CLF inequality:

$$\begin{aligned} \dot{V}(x, -kx) + l(x, -kx) &= px\dot{x} + \frac{1}{2}qx^2 + \frac{1}{2}ru^2 = px(x^3 - kx) + \frac{1}{2}qx^2 + \frac{1}{2}rk^2x^2 \\ &= \underbrace{\left(px^2 - pk + \frac{1}{2}q + \frac{1}{2}rk^2 \right)}_{= (*)} x^2 \stackrel{!}{\leq} 0 \end{aligned}$$

Since $x^2 \geq 0$, it has to hold that $(*) \leq 0$. Calculation of the values for x :

$$(*) = px^2 - pk + \frac{1}{2}q + \frac{1}{2}rk^2 \leq 0 \quad \Rightarrow \quad x^2 \leq \frac{pk - \frac{1}{2}q - \frac{1}{2}rk^2}{p}$$

Thus the CLF inequality is fulfilled for all

$$|x| \leq \sqrt{\frac{pk - \frac{1}{2}q - \frac{1}{2}rk^2}{p}}. \quad (13)$$

Thereby, $pk - \frac{1}{2}q - \frac{1}{2}rk^2 \geq 0$ needs to hold. Calculate interval boundaries from $pk - \frac{1}{2}q - \frac{1}{2}rk^2 = 0$:

$$k_{1,2} = \frac{-p \pm \sqrt{p^2 - qr}}{-r} = \frac{p \pm \sqrt{p^2 - qr}}{r}.$$

It follows that $p \geq \sqrt{qr}$ must hold, since otherwise no $k \in \mathbb{R}$ would exist for which the CLF inequality is satisfied.

Calculation of the parameter β :

$$V(x) = \frac{1}{2}px^2 \leq \frac{1}{2}(pk - \frac{1}{2}q - \frac{1}{2}rk^2)$$

$$\text{thus } \beta = \frac{1}{2}(pk - \frac{1}{2}q - \frac{1}{2}rk^2).$$

(4 P) * c) Calculate the Gâteaux derivative of $J(x)$:

$$\begin{aligned} \delta J(\mathbf{x}; \mathbf{y}) &= \lim_{\epsilon \rightarrow 0} \frac{J(\mathbf{x} + \epsilon \mathbf{y}) - J(\mathbf{x})}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \int_0^T (x + \epsilon y)^2 - x^2 dt = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \int_0^T x^2 + 2x\epsilon y + \epsilon^2 y^2 - x^2 dt \\ &= \lim_{\epsilon \rightarrow 0} \int_0^T 2xy + \epsilon y^2 dt = \int_0^T 2xy dt \end{aligned}$$

or

$$\begin{aligned} \delta J(\mathbf{x}; \mathbf{y}) &= \left. \frac{dJ(\mathbf{x} + \epsilon \mathbf{y})}{d\epsilon} \right|_{\epsilon=0} \\ &= \int_0^T \left. \frac{d}{d\epsilon} (x + \epsilon y)^2 dt \right|_{\epsilon=0} = \int_0^T 2(x + \epsilon y)y dt \Big|_{\epsilon=0} = \int_0^T 2xy dt \end{aligned}$$

$\Rightarrow$ The Gâteaux derivative exists for all points y .